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Experiment Name: Array Functions in Data Science

```
import numpy as np
array=np.random.randint(1,100,9)
array
```

```
array([ 6, 11, 49, 88, 98, 80, 24, 43, 12], dtype=int32)
```

```
np.sqrt(array)
```

```
array([2.44948974, 3.31662479, 7.        , 9.38083152, 9.89949494,
       8.94427191, 4.89897949, 6.55743852, 3.46410162])
```

```
array.ndim
```

```
1
```

```
new_array=array.reshape(3,3)
```

```
new_array
```

```
array([[ 6, 11, 49],
       [88, 98, 80],
       [24, 43, 12]], dtype=int32)
```

```
new_array.ndim
```

```
2
```

```
new_array.ravel()
```

```
array([ 6, 11, 49, 88, 98, 80, 24, 43, 12], dtype=int32)
```

```
newm=new_array.reshape(3,3)
```

```
newm
```

```
array([[ 6, 11, 49],
       [88, 98, 80],
       [24, 43, 12]], dtype=int32)
```

```
newm[2,1:3]
```

```
array([43, 12], dtype=int32)
```

```
newm[1:2,1:3]
```

```
array([[98, 80]], dtype=int32)
```

```
new_array[0:3,0:0]
```

```
array([], shape=(3, 0), dtype=int32)
```

```
new_array[0:2,0:1]

array([[ 6],
       [88]], dtype=int32)

new_array[0:3,0:1]

array([[ 6],
       [88],
       [24]], dtype=int32)

new_array[1:3]

array([[88, 98, 80],
       [24, 43, 12]], dtype=int32)
```